

Mohamed Abdelrahman

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EDUCATION

University of Houston

Bachelor of Science in Computer Science - Major GPA: 3.80

Expected Graduation: Dec 2026

Houston, TX

EXPERIENCE

Course Facilitator

Department of Computer Science - University of Houston

Jan 2026 – Present

Houston, TX

- Preparing coding problems and instructional material to facilitate COSC 3320: Algorithms and Data Structures for undergraduate students

Software Engineer (Consultant)

bld.ai

Nov 2022 – Apr 2023

Remote

- Developed a full-stack platform using React and Django to match families for playdate arrangements
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month using a two-week sprint cycle
- Designed and configured role-based permission groups to enforce to ensure data integrity and controlled access

LEADERSHIP

Director of Coding Competitions

IEEE-NSM - University of Houston

Jan 2025 – Present

- Prepared and hosted workshops on data structures and algorithms in C++, and curated competitive programming competitions to strengthen undergraduate students' algorithmic problem-solving skills
- Organized practice sessions for the ICPC team by selecting problems and arranging virtual contest sessions, contributing to the team's 1st place finish at UH Site
- Coordinated with students and faculty on ICPC hosting logistics at UH, organizing scheduling shifts for 20 volunteers to ensure successful contest execution

Code[Coogs] - University of Houston

- Organized and hosted 9 competitive programming events on Kattis, curating problem sets focused on data structures, algorithms, and complexity analysis for 30+ students

AWARDS

Golden Medalist, ICPC South Regional Contest - UH Site

University of Houston

Nov 2025

- Secured 1st place at UH Site (16th place regionally) in the Division 1 ICPC South Regional Contest by solving 4 problems using C++ and advanced algorithmic techniques

PROJECTS

Echo Journal (AI Voice Diary) | *Cloudflare Workers, Python (Durable Objects), Llama 3, React*

- Architected a serverless voice journaling app using Cloudflare Durable Objects to consolidate Whisper transcription, Llama 3 inference, and SQL state within isolated, low-latency user instances
- Engineered a persistent conversation context system that updates prompt history for coherent multi-turn dialogue, coupled with a privacy-first auto-deletion mechanism for inactive user data

C++ Ordered Map Library | *C++17, CMake, Google Test, Valgrind, GitHub Actions*

- Developed a high-performance ordered map data structure combining a hash table with a custom doubly linked list to achieve $O(1)$ lookups while preserving insertion order for LRU cache simulations
- Implemented "node splicing" logic to enable fast Move-To-Front operations without expensive memory reallocations, validating zero memory leaks via Valgrind and reliability with Google Test

TallyCart | *Python, Django, PostgreSQL, Docker, Celery, OpenCV*

- Built a receipt-to-data pipeline combining OpenCV for image preprocessing and LLM-based parsing to extract structured product data from non-standardized receipt formats
- Designed an asynchronous architecture using Celery workers to handle heavy OCR tasks in the background, implementing a normalization layer to standardize units for cross-store price comparison

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart, C#

Frameworks: Django, React, NextJS, Flutter, Unity

Tools: Git, Docker, Postman, Coverage, GitHub Actions, VS Code, Ubuntu Linux, Bash